



Composite Pattern

All things you should know about Composite.



What?

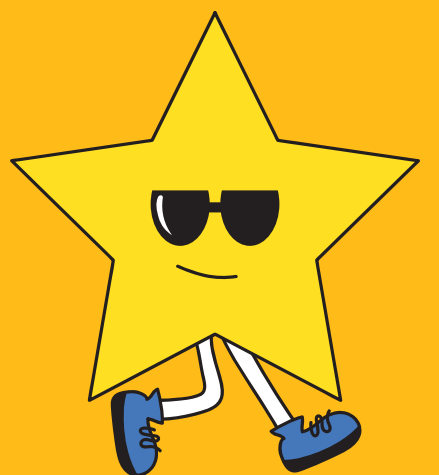
It belongs to the structural
category.

Compose objects into tree
structures to represent part
or whole of hierarchies.

Composite lets clients treat
individual objects and
compositions of objects
seamlessly.



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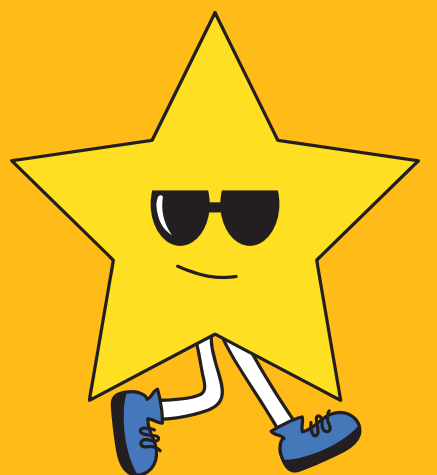


Real-World Scenario

Imagine you're building a graphical editor.
Users can create basic shapes like Text, Circle,
etc., and combine them into complex images.
A basic solution might involve one class for
simple objects and another for containers. But
this forces clients to handle each differently,
increasing complexity.



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Solution

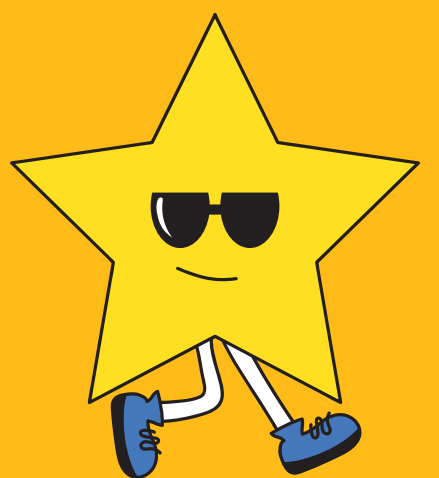
Composite pattern helps us to recursive composition so that clients are not required to make this distinction. This pattern has an abstract class for both primitives and their containers.

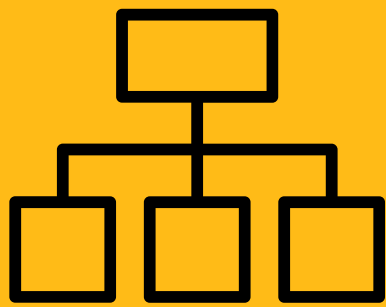
For instance, in the graphics system this abstract named Graphic which declares operation like Draw for graphical objects. It also declares operations that all composite objects share, such as Get Child, Add, Remove.

The Picture class defines a list of graphic objects and implements draw to call draw on its children. In addition, the picture implements child-related operations. Since the picture interface inherits from the Graphic interface, picture objects can compose other pictures recursively.

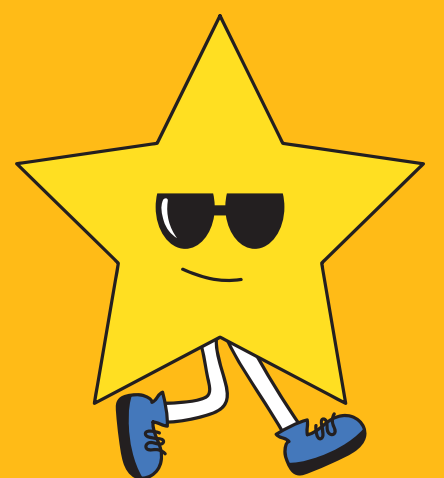
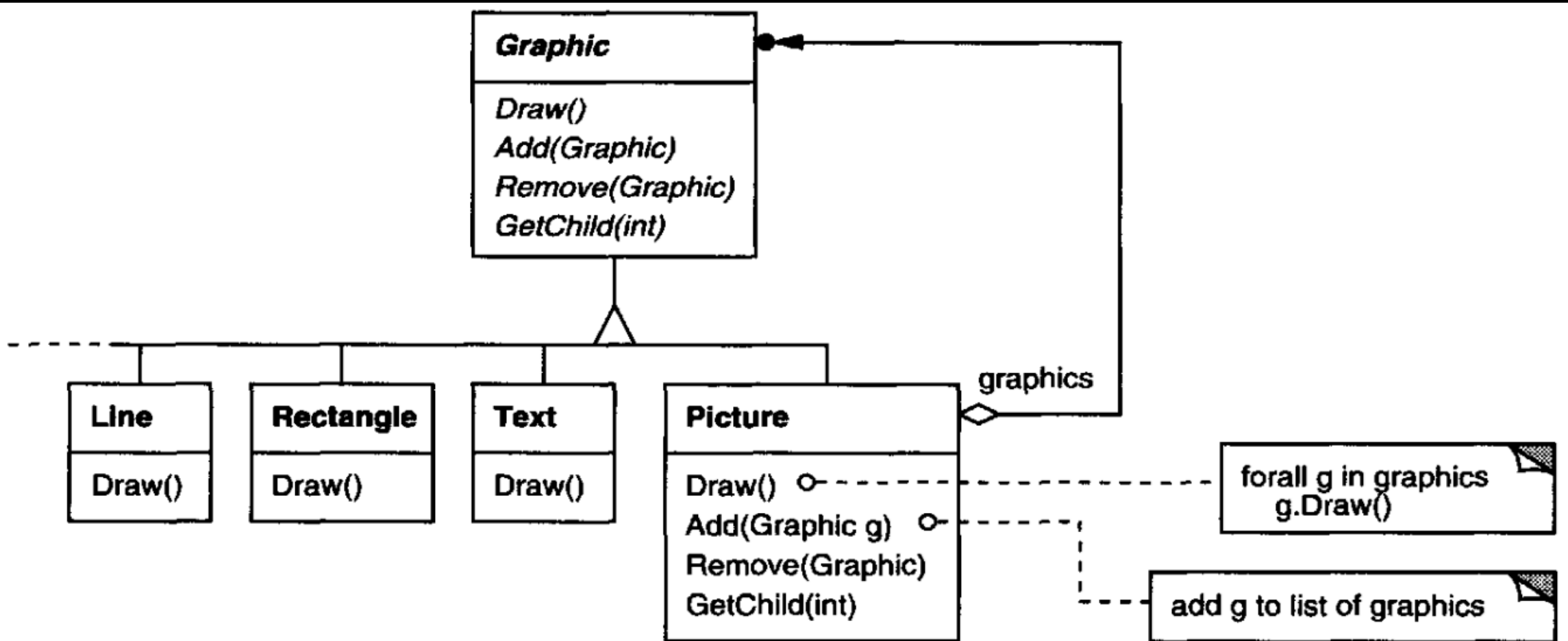


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Structure





Usability



1. You need to design a component where individual objects and groups of objects can be treated uniformly.
2. Your goal is to allow clients to interact with object compositions and individual objects uniformly.



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